Assignment No. 12

Comparative study: Under Relaxation Factor
Vs
Pseudo Transient Method

13.08.2026

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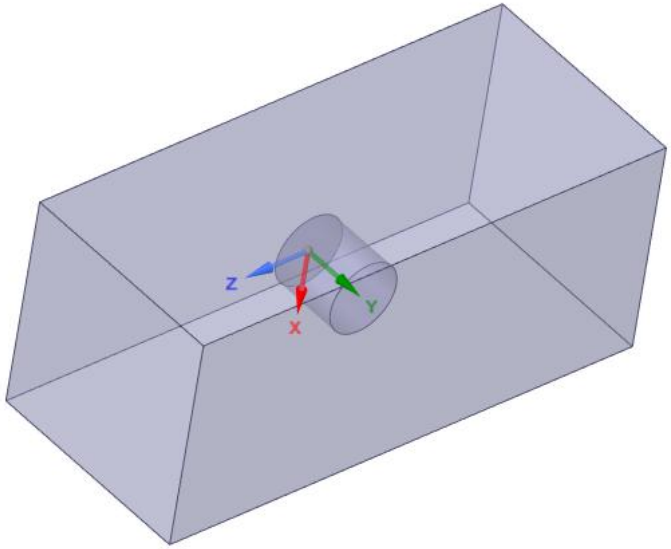
Objective

- To compare result and residual behavior for URF and PT methods

Problem Statement

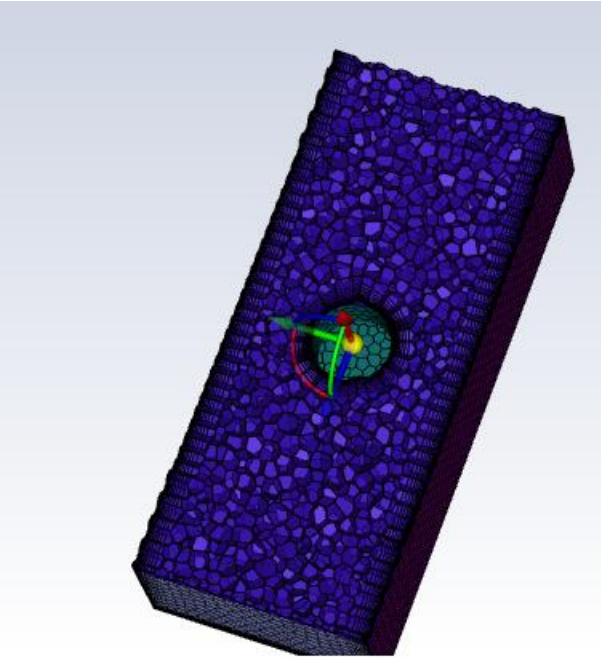
- Water flow over a pipe of 1m diameter. Inlet velocity: 50 m/s

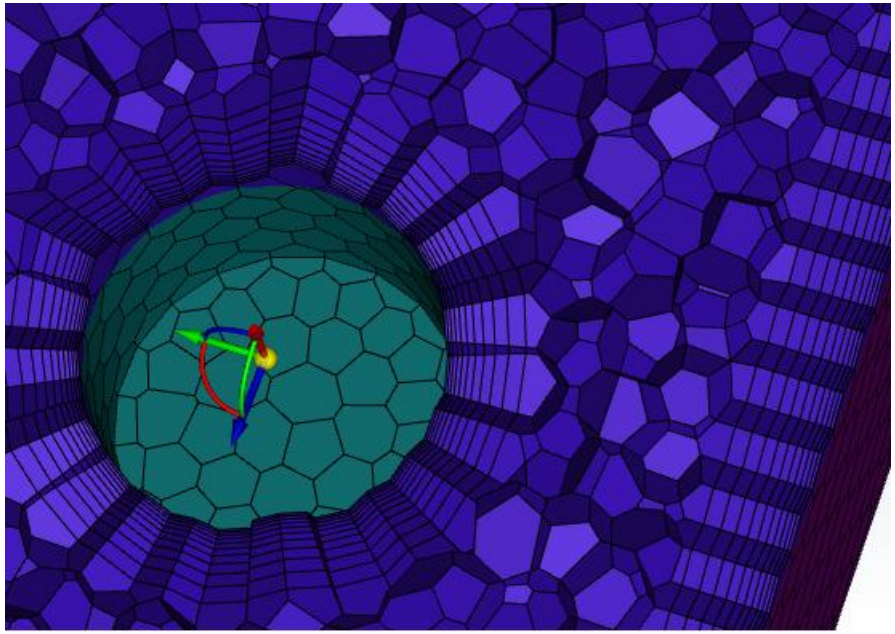
Geometry



Blocking

- 10 boundary layers were added for better capture of velocity profile

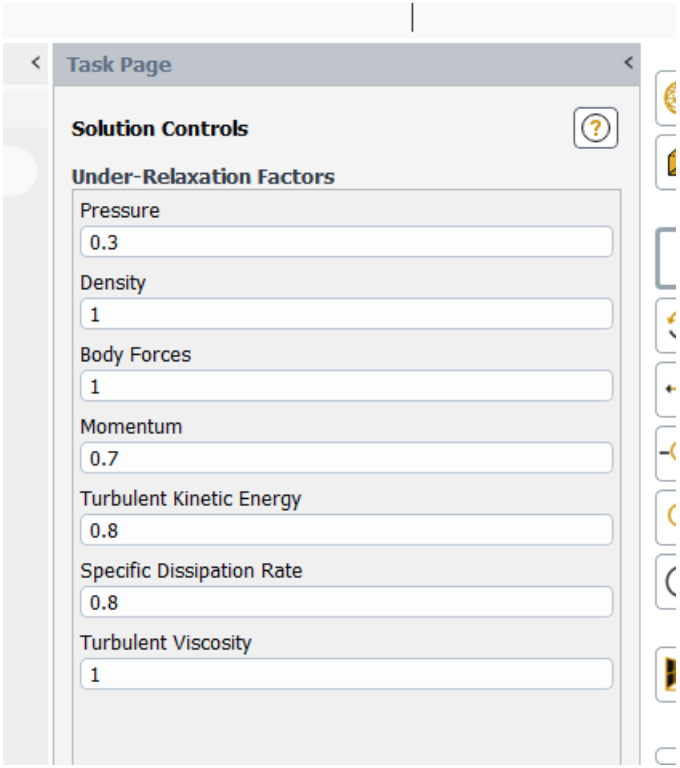


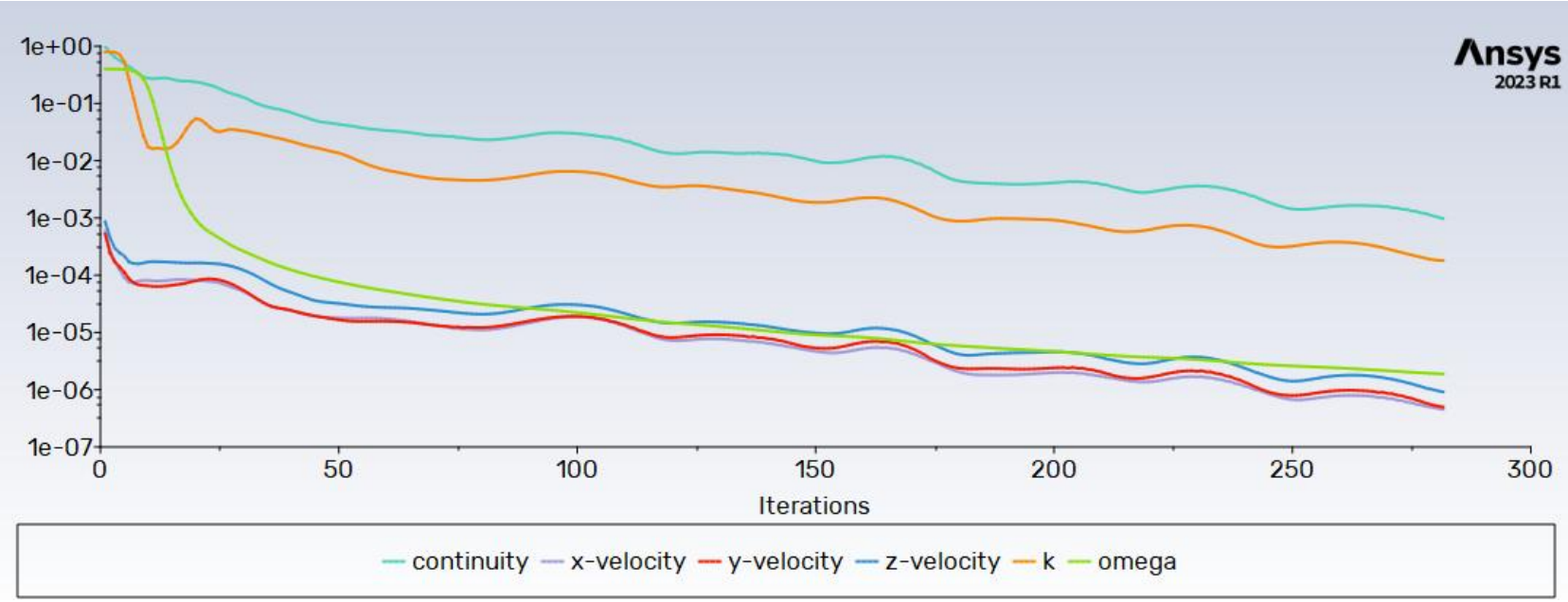


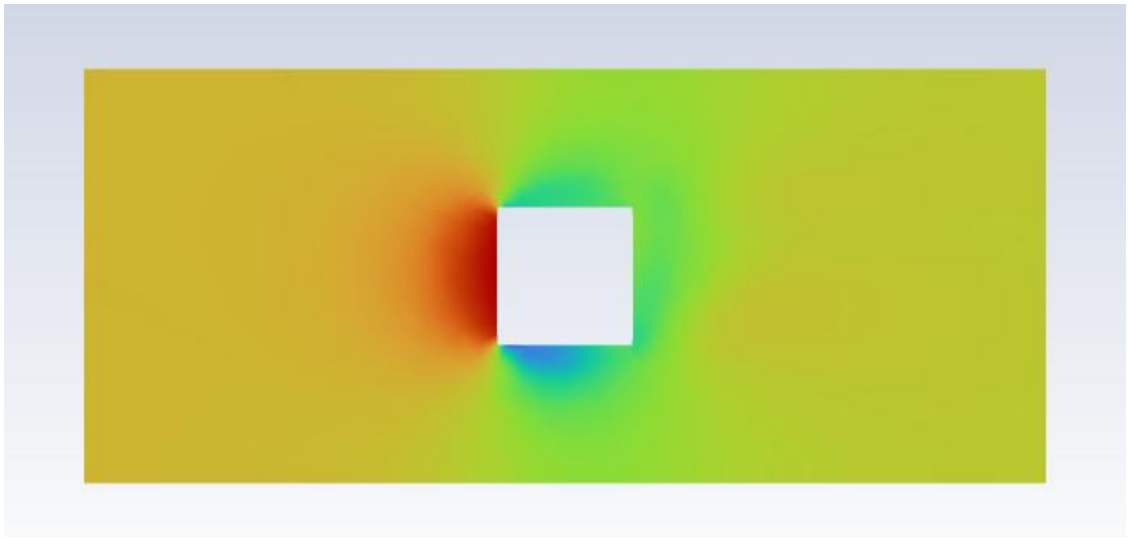
CFD Setting

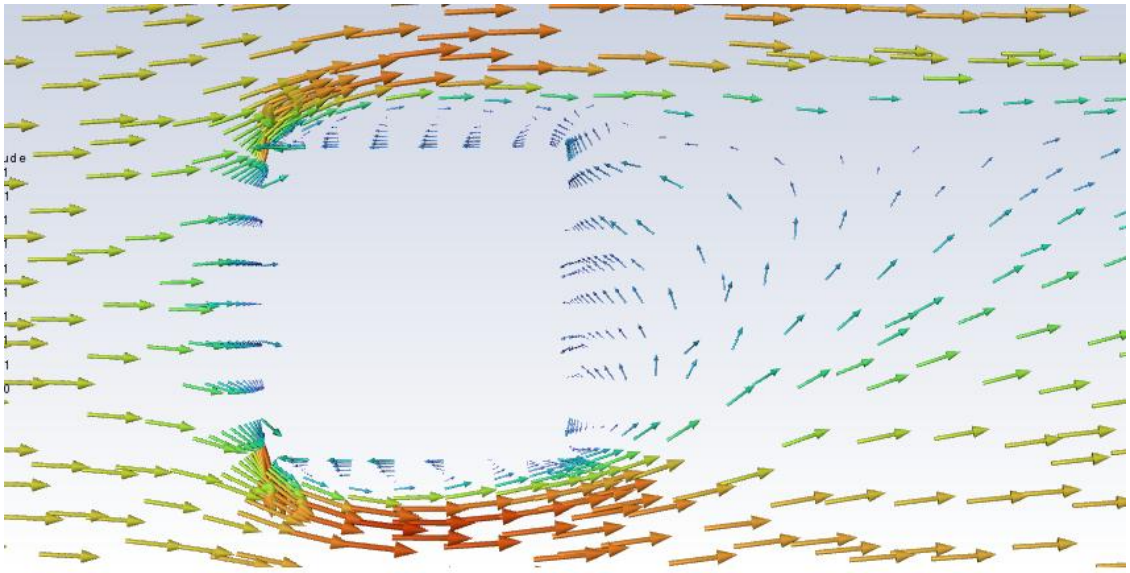
- Element count: 47,000
- Steady Sate
- No gravity
- No energy equation
- Turbulence Model: SST K-omega(2 Eqn)
- Velocity inlet = 50 m/s and pressure outlet
- Material: Liquid Water

Results(URF)

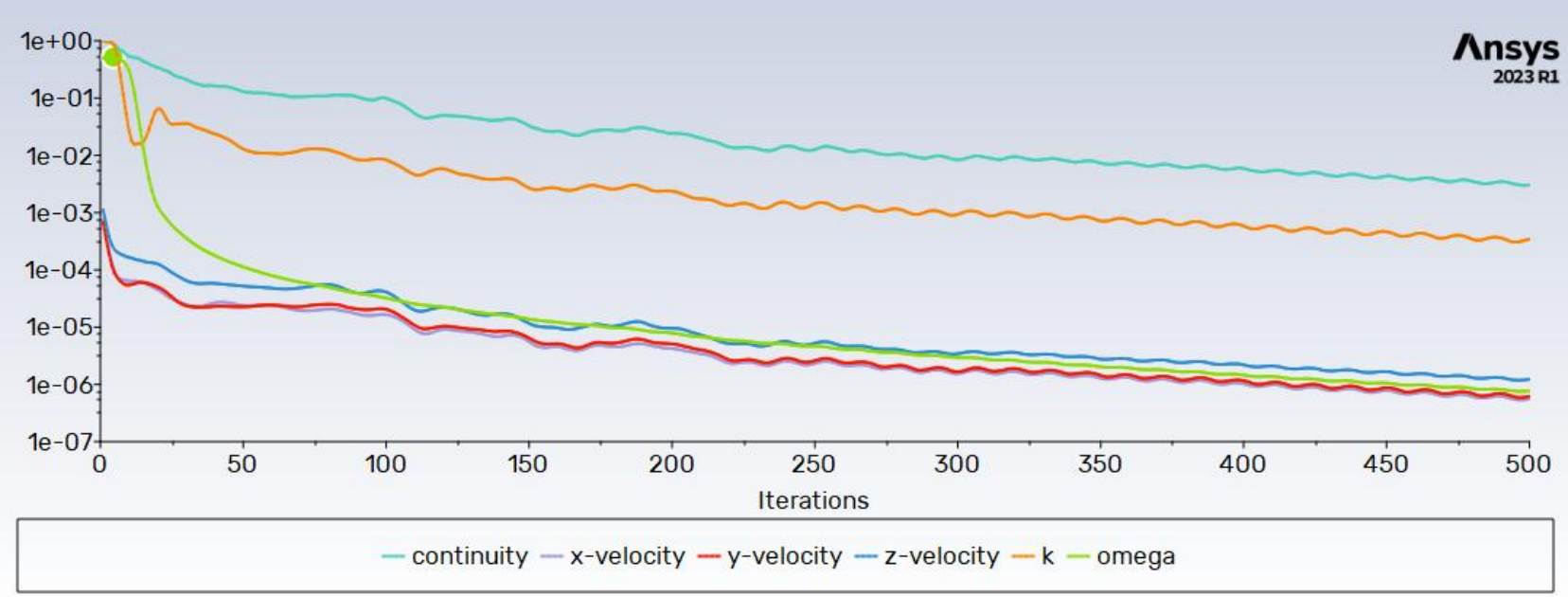

Task Page
Solution Controls
Under-Relaxation Factors
Pressure
0.3
Density
1
Body Forces
1
Momentum
0.7
Turbulent Kinetic Energy
0.8
Specific Dissipation Rate
0.8
Turbulent Viscosity
1

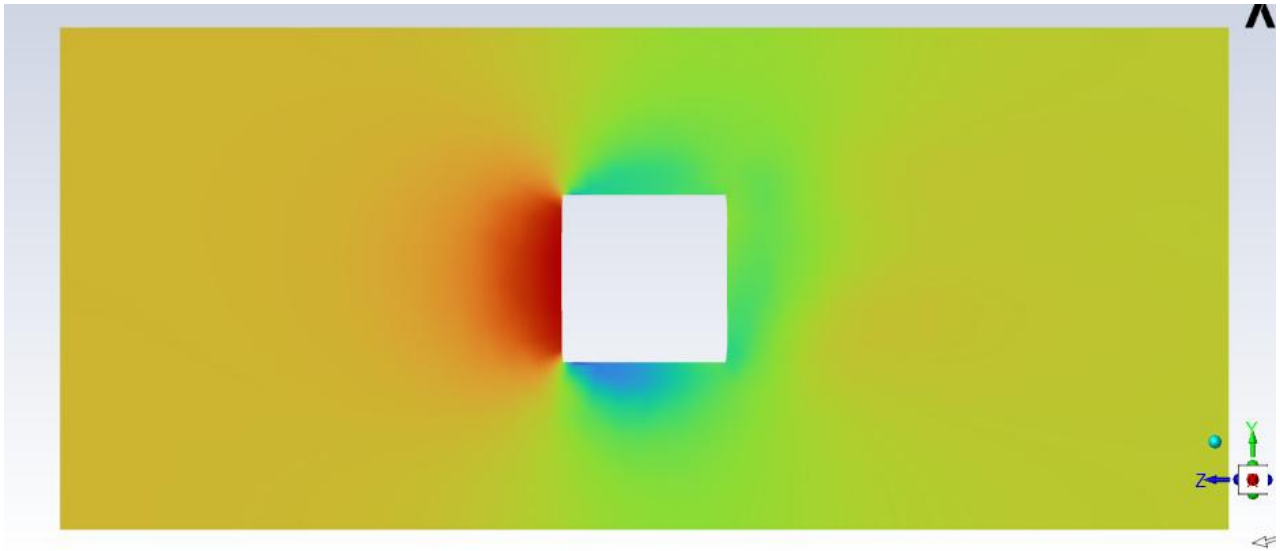

Ansys
2023 R1
1e+00
1e-01
1e-02
1e-03
1e-04
1e-05
1e-06
1e-07
0
50
100
150
200
250
300
Iterations
continuity
x-velocity
y-velocity
z-velocity
k
omega

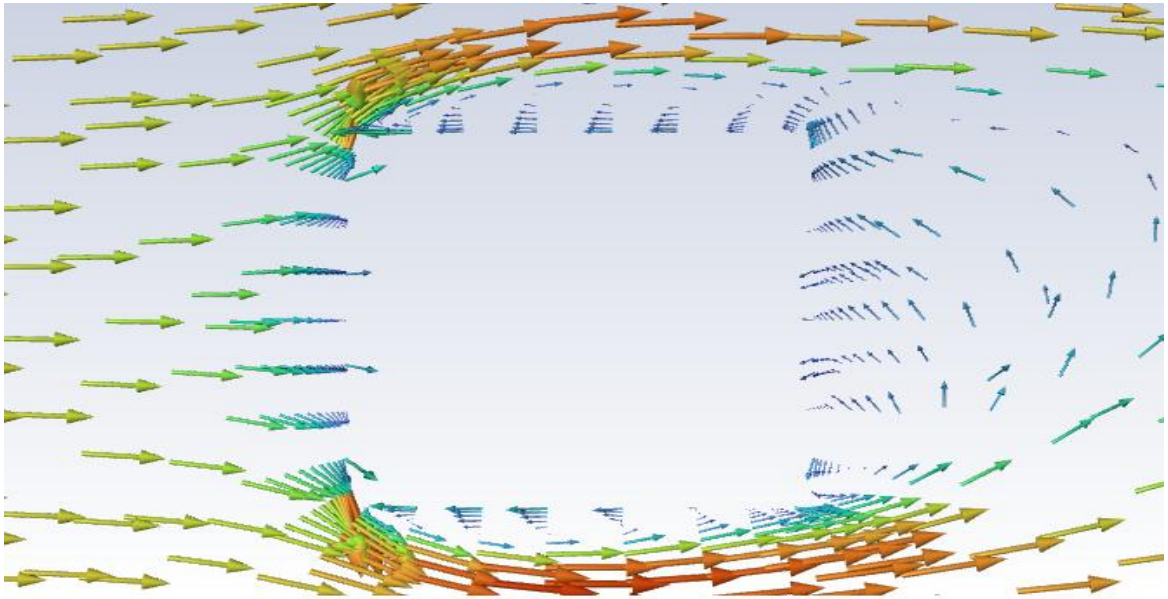




Results(PT method)




Y
Z



Conclusion

- URF converged before hitting 300 iterations. PT didn't converge at even 500 iterations. Also the residual graph suggests that URF was much more stable for this particular scenario.